

AI Capstone Project –Guidelines

1. What is this project?

This capstone project is not a demo. It is a real AI application that should be strong enough to be added to your portfolio. You are expected to build a real solution to a real problem and deploy it as a working system.

2. General Project Setup

- Team size: 5 students
- Duration: 4 weeks
- Project type: Full-stack AI application
- Output: Production-ready MVP

3. Core Project Requirements

- Every project must include:
 - A real problem
 - A working full-stack application
 - Git version control
 - Cloud deployment
 - Clear technical and user documentation

4. Mandatory AI Capabilities

- Your project must include:
 - LLM integration using a real provider
 - RAG to retrieve answers from documents
 - Agentic AI that can reason and use tools

5. LLM Integration

Use a real LLM (OpenAI, Gemini, Claude, or similar). Focus on proper usage, prompt design, secure API keys, and meaningful interaction.

6. RAG (Retrieval-Augmented Generation)

Your system must use documents or data sources. The AI should retrieve information from these sources before answering.

7. Agentic AI

Agents should reason, use tools, and make decisions. At least one agent is required, with a clear role.

8. Additional Areas

- Multimodal AI (choose at least two)
- Orchestration framework (choose one)
- Security & basic compliance
- Monitoring and evaluation
- UX requirements
- Cost awareness and scalability
- Testing and DevOps

9. Evaluation Levels

- Bronze: Working MVP
- Silver: Production-ready
- Gold: Industry-level

10.Final Deliverables

- GitHub repository
- Live deployed app
- Documentation and diagrams
- Demo video
- Presentation
- Evaluation report

11.Key Philosophy

This is not a demo. Quality matters more than quantity. This project represents you as a developer.